

JEE MAIN REVISION MASTER: RAY OPTICS 2026 5 QUESTIONS 8.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Reflection at Spherical Mirrors & Mirror Formula	19 PYQs	Mirrors
Very High	Refraction at Interfaces, Snell's Law & TIR	18 PYQs	Refraction
High	Lens Maker's Formula & Optical Instruments	14 PYQs	Lenses & Instruments
Moderate	Prism Dispersion & Minimum Deviation	7 PYQs	Prisms

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Reflection at Spherical Mirrors & Mirror Formula	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $\text{quad } m = -\frac{v}{u} = \frac{f}{f - u}$, $\text{quad } f = \frac{R}{2}$
Refraction at Interfaces, Snell's Law & TIR	$n_1 \sin i = n_2 \sin r$, $\text{quad } \sin \theta_c = \frac{n_2}{n_1}$; ($n_1 > n_2$), $\text{quad } d' = \frac{d}{\mu}$, $\text{quad } \text{Shift} = t \left(1 - \frac{1}{\mu} \right)$
Lens Maker's Formula & Optical Instruments	$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$, $\text{quad } P = \frac{1}{f}$, $\text{quad } M_{\text{telescope}} = \frac{f_o}{f_e}$, $\text{quad } M_{\text{microscope}} = \frac{L}{f_o} \frac{D}{f_e}$
Prism Dispersion & Minimum Deviation	$\delta = i + e - A$, $\text{quad } \mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin(A/2)}$, $\text{quad } \delta \approx (\mu - 1)A$; (thin prism)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Reflection at Spherical Mirrors & Mirror Formula (Very High Yield — 19 PYQs)

Core Principle: Image formation by concave and convex mirrors, Cartesian sign convention, and linear/longitudinal magnification.

■ **Caution / JEE Pitfall:** Concave mirror has negative focal length ($f < 0$); convex mirror has positive focal length ($f > 0$).

■ **Speed Trick:** Virtual image in a concave mirror is ALWAYS magnified ($|m| > 1$), while in a convex mirror it is ALWAYS diminished ($|m| < 1$).

Representative Source PYQs:

- [2026 Shift] Q#4 (MCQ): "The direction of a ray of light incident on a concave mirror is shown by PQ, while direction in which the ray would travel after reflection ..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An equiconvex lens of radius of curvature 14 cm is made up of two different materials. Left half and right half of vertical portion is made ..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The final image formed by an astronomical telescope is..." [Ans: N/A]

3.2 Refraction at Interfaces, Snell's Law & TIR (Very High Yield — 18 PYQs)

Core Principle: Snell's law, apparent depth shift in transparent media, and total internal reflection condition ($i \geq \theta_c$ from denser to rarer).

■ **Caution / JEE Pitfall:** Total internal reflection can ONLY occur when light travels from an optically DENSER medium to a RARER medium.

■ **Speed Trick:** Area of circle through which light emerges from fish/source at depth h : $A = (\pi h^2 / \mu^2 - 1)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The critical angle for a monochromatic light going from medium A to medium B is θ . If the speed of light in medium A is V , then t ..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "The incorrect statement about refractive index for a pair of media is..." [Ans: N/A]

3.3 Lens Maker's Formula & Optical Instruments (High Yield — 14 PYQs)

Core Principle: Thin lens refraction, Lens Maker's equation, lens power in Dioptres ($P = 1/f$), and magnifying power of optical devices.

■ **Caution / JEE Pitfall:** When a lens is immersed in a liquid of refractive index $\mu_l > \mu_{\text{(glass)}}$, its converging/diverging nature REVERSES (convex acts concave).

■ **Speed Trick:** When a lens is cut vertically in half, each half retains the SAME focal length f . When cut horizontally, focal length remains f , but image brightness is halved.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "A convex lens has power P . It is cut into two halves along its principal axis. Further one piece (out of two halves) is cut into two halves ..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The image formed by an objective lens of a compound microscope is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An equiconvex lens of radius of curvature 14 cm is made up of two different materials. Left half and right half of vertical portion is made ..." [Ans: N/A]

3.4 Prism Dispersion & Minimum Deviation (Moderate Yield — 7 PYQs)

Core Principle: Ray refraction through triangular prisms, condition for minimum deviation ($i = e$, $r_1 = r_2 = A/2$), and angular dispersion.

■ **Caution / JEE Pitfall:** At minimum deviation, the refracted ray inside an equilateral prism passes strictly PARALLEL to the base.

■ **Speed Trick:** Condition for no emergent ray (always TIR on second face): $A > 2\theta_c$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is..." **[Ans: N/A]**
- **[2026 Shift] Q#1 (MCQ):** "With reference to the figure shown below, match the following: List-I List-II (a) Angle of reflection (i) 60° (b) Value of α ..." **[Ans: A]**
- **[2025 Shift] Q#1 (MCQ):** "If r and r^1 denotes the angles inside the prism having angle of prism 50° considering that during interval of time from $t=0$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is

- (A) $\frac{\sqrt{3}}{2}$
- (B) $\frac{3}{2}$
- (C) $\sqrt{3}$
- (D) $\sqrt{2}$

Correct Answer: None

Solution Analysis: For an equilateral prism, the angle of prism is $A=60^\circ$. From the graph of angle of deviation δ versus angle of incidence i , the lowest value of deviation is the minimum deviation. Here, the graph shows $\delta_m=60^\circ$. The refractive index μ of the prism material is given by $\mu = \frac{\sin(A + \delta_m/2)}{\sin(A/2)}$. Now put $A=60^\circ$ and $\delta_m=60^\circ$ in the formula. $\mu = \frac{\sin(60^\circ + 60^\circ/2)}{\sin(60^\circ/2)} = \frac{\sin(90^\circ)}{\sin(30^\circ)} = \frac{1}{1/2} = 2$ Using $\sin 60^\circ = \frac{\sqrt{3}}{2}$ an

Benchmark Problem 2: KCET 2026 (Q#1)

With reference to the figure shown below, match the following: List-I List-II (a) Angle of reflection (i) 60° (b) Value of α (ii) 120° (c) Angle of deviation (iii) 30°

- (A) a-i, b-ii, c-iii **[Correct]**
- (B) a-ii, b-i, c-iii
- (C) a-iii, b-i, c-ii
- (D) a-iii, b-ii, c-i

Correct Answer: A

Solution Analysis: From the figure, the angle of incidence is $i=30^\circ$. By the law of reflection, the angle of reflection is equal to the angle of incidence. So, $r=30^\circ$. This matches (a) with (iii). The normal is perpendicular to the mirror surface. So the angle between the normal and the mirror is 90° . Here, $r+\alpha=90^\circ \Rightarrow 30^\circ+\alpha=90^\circ \Rightarrow \alpha=60^\circ$ So, (b) matches with (i). The angle of deviation is given by $\delta=180^\circ-(i+r) \Rightarrow \delta=180^\circ-(30^\circ+30^\circ) \Rightarrow \delta=120^\circ$ So,

Benchmark Problem 3: KCET 2026 (Q#4)

The direction of a ray of light incident on a concave mirror is shown by PQ, while direction in which the ray would travel after reflection is shown by four rays marked as A, B, C and D as shown in the figure. Which of the four rays correctly shows the direction of the reflected ray?

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: For a concave mirror, there is an important rule: a ray of light which is parallel to the principal axis reflects and then passes through the principal focus, F . In the figure, the incident ray PQ is parallel to the principal axis of the concave mirror. So after reflection, this ray must pass through the focus F . Among the given rays, only ray C passes through the principal focus F . So, the correct reflected ray is C.

Benchmark Problem 4: KCET 2025 (Q#4)

The image formed by an objective lens of a compound microscope is

- (A) Real and diminished
- (B) Real and enlarged
- (C) Virtual and enlarged
- (D) Virtual and diminished

Correct Answer: None

Solution Analysis: The objective lens of a compound microscope produces a real, inverted and magnified image of the object. So the correct choice is: Option B: Real and enlarged